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Topic 7: Animal Coordination, Control and Homeostasis

Edexcel GCSE (9-1) Biology (1BI0) — Paper 2

Your body must coordinate its activities so that cells, tissues and organs work together. Two systems do this: the **nervous system** (covered in Topic 2) which uses fast electrical impulses, and the **endocrine system** which uses chemical messengers called **hormones**. Topic 7 focuses on the endocrine system and how it contributes to **homeostasis** — the maintenance of constant internal conditions (blood glucose, body temperature, water balance).

> **Homeostasis** is the regulation of the internal conditions of a cell or organism to maintain optimum conditions for function, in response to internal and external changes. It typically involves **negative feedback**: a change from the set point triggers responses that restore the set point.

7.1 The Endocrine System

> The **endocrine system** is a collection of **glands** that secrete chemical messengers called **hormones** directly into the **blood**. Blood carries the hormone to **target organs**, where it produces an effect by binding to specific receptors on target cells.

Key features of hormonal communication:

- **Slow** to take effect — blood has to carry the hormone.
- **Long-lasting** — hormones persist in the blood until broken down (often by the liver).
- **Wide-reaching** — a single hormone can affect many organs at once.

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- **Specific** — only cells with the correct receptor respond.

Nervous vs Endocrine Systems

Feature	Nervous System	Endocrine System
Type of signal	Electrical impulses along neurones	Chemical hormones in the blood
Speed of response	Very fast (milliseconds)	Slower (seconds to minutes)
Duration of response	Short-lived	Long-lasting
Area affected	Very precise — one effector (e.g. a muscle)	Can be widespread — any cell with the receptor
Transmission path	Neurones (axons)	Bloodstream
Example response	Pulling hand off a hot plate (reflex)	Puberty, blood glucose control, menstrual cycle

Exam Tip: If an exam asks you to compare nervous and hormonal control, always cover: (1) signal type, (2) speed, (3) duration, (4) how precisely targeted. Use the word "target organ" when describing hormone action.

7.2 Major Endocrine Glands

You need to know the position and function of the main endocrine glands in the body.

Gland	Location	Main hormones
Pituitary ("master gland")	Base of the brain	FSH, LH, ADH, growth hormone — many control other glands

Thyroid	Neck (in front of the windpipe)	Thyroxine
Adrenal glands	One on top of each kidney	Adrenaline
Pancreas	Just below the stomach	Insulin, glucagon
Ovaries (in females)	Lower abdomen	Oestrogen, progesterone
Testes (in males)	In the scrotum	Testosterone

The **pituitary gland** is called the "master gland" because it secretes hormones that act on other endocrine glands (e.g. FSH acts on the ovaries, TSH acts on the thyroid) — so it coordinates much of the endocrine system.

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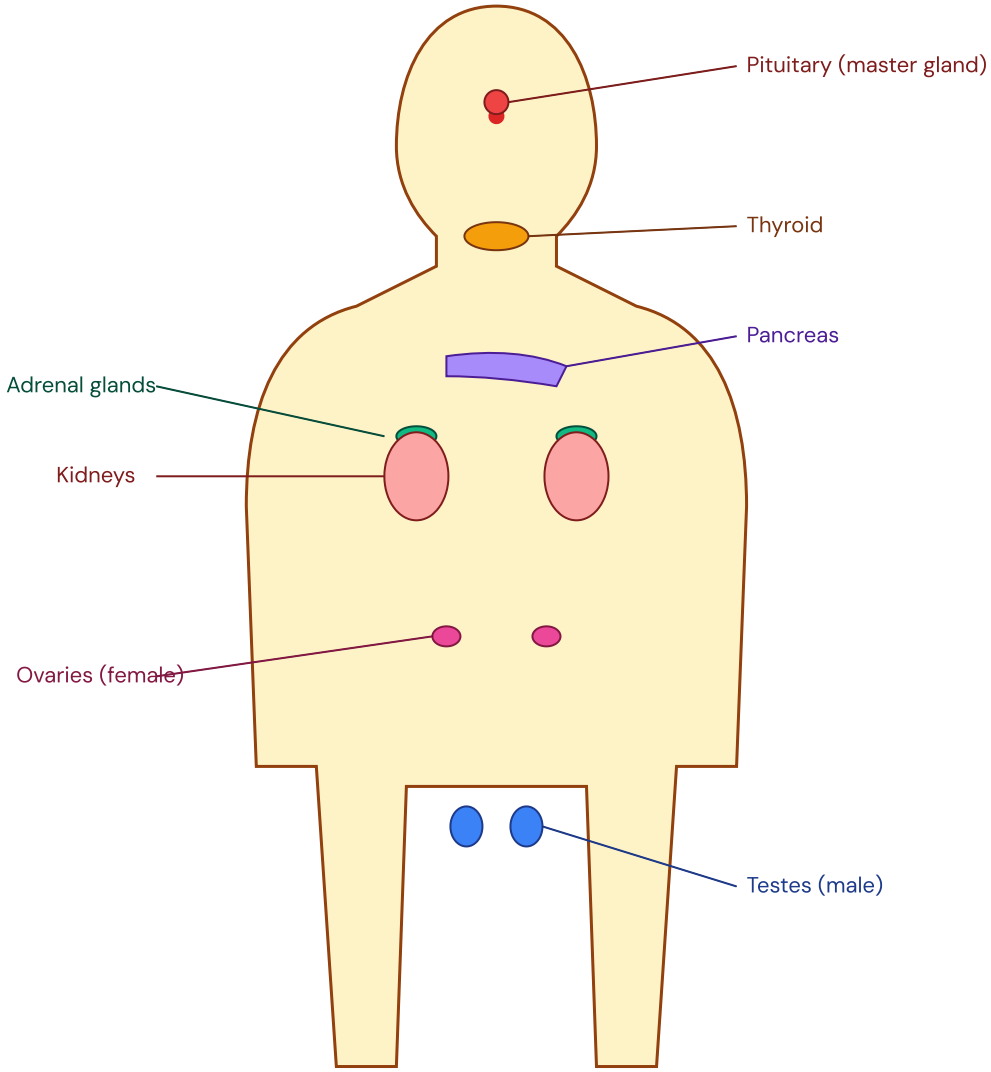
Major Endocrine Glands



7.3 Key Hormones and Their Functions

Gland	Hormone	Target & function
Pituitary	FSH (follicle-stimulating hormone)	Ovaries — stimulates follicle development & oestrogen release
	LH (luteinising hormone)	Ovaries — triggers ovulation at mid-cycle

	ADH (antidiuretic hormone)	Kidneys — increases water reabsorption (water balance)
	Growth hormone	Most tissues — stimulates growth in children/adolescents
Thyroid	Thyroxine	Most cells — controls basal metabolic rate, growth, heat production
Adrenal glands	Adrenaline	Heart, muscles, liver — "fight-or-flight" response
Pancreas	Insulin	Liver & muscle — lowers blood glucose (stores it as glycogen)
	Glucagon	Liver — raises blood glucose (breaks glycogen down to glucose)
Ovaries	Oestrogen	Uterus & pituitary — builds uterine lining, controls menstrual cycle, female secondary sexual characteristics
	Progesterone	Uterus — maintains uterine lining during cycle/pregnancy
Testes	Testosterone	Reproductive organs & body — sperm production, male secondary sexual characteristics

Adrenaline — the Fight-or-Flight Hormone

Adrenaline is released by the **adrenal glands** when the body is under stress (fear, excitement, danger). Its effects prepare the body for rapid action:

- **Heart rate increases** — more oxygen and glucose to muscles per minute.
- **Breathing rate increases** — more oxygen taken in.
- **Pupils dilate** — more light enters the eye, better vision.
- **Blood vessels to muscles dilate, those to gut constrict** — blood is redirected to muscles.

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- **Liver releases glucose from glycogen** — more respiratory substrate for muscles.
- **Sweating** — prepares to lose heat from extra respiration.

> **Exam Tip:** Adrenaline effects and vital statistics (heart rate, breathing, pupil size) are popular 4- and 6-mark questions. Always link each effect back to "more oxygen and glucose reach the muscles faster" so they can respire and power action.

Thyroxine and Negative Feedback

Thyroxine from the thyroid controls the **basal metabolic rate** — the rate at which cells respire and use energy at rest. Its release is itself controlled by negative feedback:

1. Pituitary releases **TSH** (thyroid-stimulating hormone).
2. TSH stimulates the thyroid to release thyroxine.
3. Rising blood thyroxine feeds back on the pituitary to inhibit TSH.
4. If thyroxine falls, inhibition weakens and TSH rises again.

7.4 Blood Glucose Regulation

Blood glucose must be kept close to a set point (~5 mmol dm^{-3} when fasting, ~7 mmol dm^{-3} after a meal). If it falls too low, cells — especially brain cells — cannot respire enough; if it rises too high it damages tissues. The **pancreas** monitors blood glucose and responds by releasing either **insulin** or **glucagon**.

Condition	Hormone released	Target	Effect on glucose
Blood glucose too high (after a meal)	Insulin (from β-cells)	Liver & muscle cells	Glucose taken up from blood; stored as **glycogen** → blood glucose falls
Blood glucose too low (between meals, exercise)	Glucagon (from α-cells)	Liver cells	Glycogen broken down to glucose; glucose released into blood → blood glucose rises

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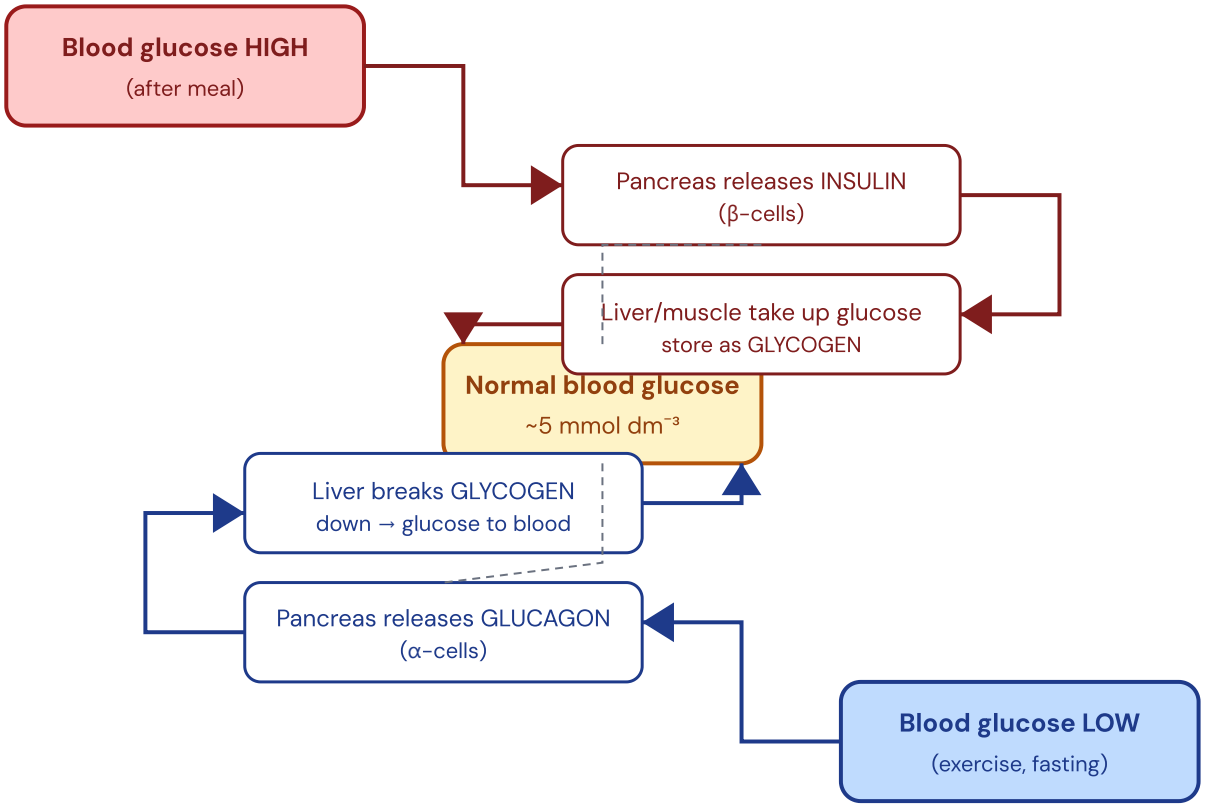
Exam Tip: Don't muddle up the words — **glucose** (the sugar in blood), **glycogen** (storage polymer in liver/muscle), **glucagon** (the hormone that raises blood glucose). Spell each one correctly!

Blood Glucose — Negative Feedback



Negative feedback: each response returns blood glucose toward the set point.

7.5 Diabetes

Diabetes is a condition in which blood glucose is not properly regulated, so it rises dangerously high after meals (**hyperglycaemia**) and cannot be brought back to normal.

Feature	Type 1 diabetes	Type 2 diabetes

Cause	Autoimmune destruction of β-cells in the pancreas — little or no insulin produced	Body cells become **resistant** to insulin; pancreas may also produce less
Age of onset	Usually childhood/adolescence	Usually adulthood (risk rises with age)
Risk factors	Genetic predisposition; not linked to lifestyle	**Obesity**, poor diet, inactivity, ethnicity, family history
Treatment	Regular **insulin injections**; monitor blood glucose; carbohydrate-controlled diet; exercise	Diet control (low sugar, low refined carbs), exercise, weight loss; later, oral drugs (e.g. metformin) or insulin
Cure?	No cure — lifelong	Often controlled / reversed by lifestyle changes in early stages

A healthy person's blood glucose rises after a meal but is brought back to the normal range within about 2 hours by insulin. In untreated Type 1 diabetes, blood glucose rises much higher after a meal and stays elevated for much longer because no insulin is available to trigger uptake. Excess glucose spills into the urine (**glycosuria**) and draws water out by osmosis, causing dehydration and thirst.

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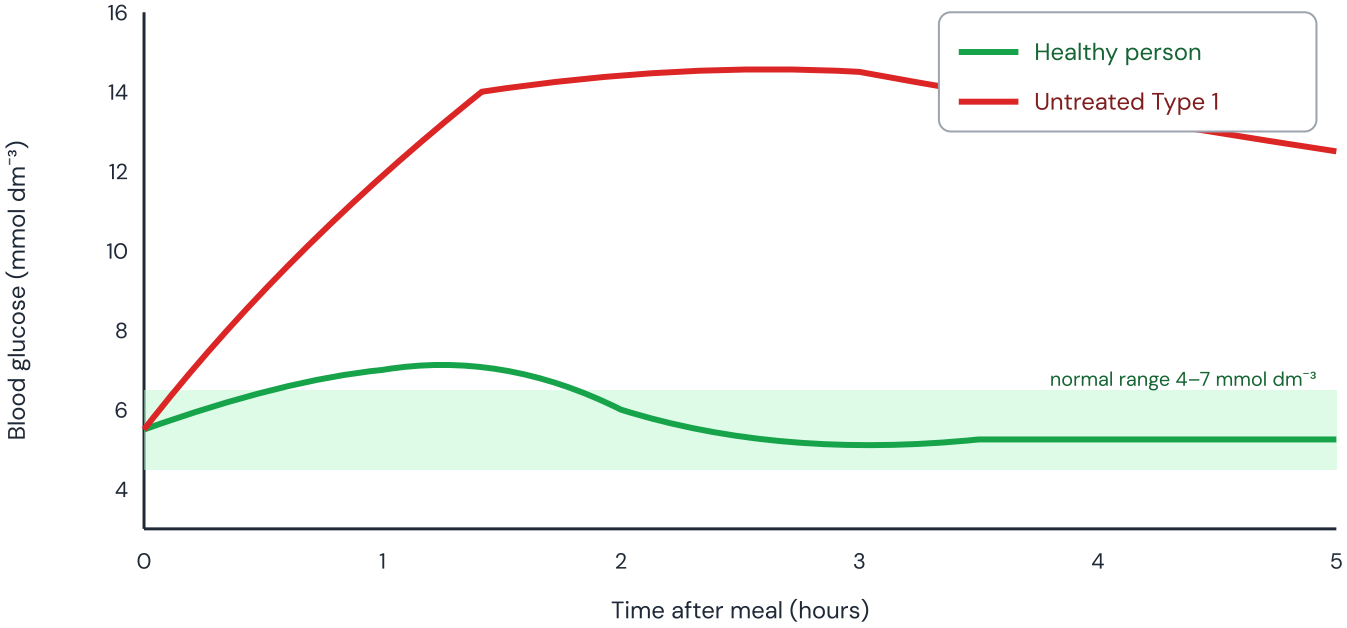
Blood Glucose after a Meal: Healthy vs Untreated Type 1



Exam Tip: A classic 6-mark question: "Explain the difference in blood glucose between a healthy person and someone with untreated Type 1 diabetes after a meal." Walk through: meal → glucose absorbed → healthy person secretes insulin → glucose taken up by liver/muscle → returns to set point. Type 1: no insulin, glucose stays high.

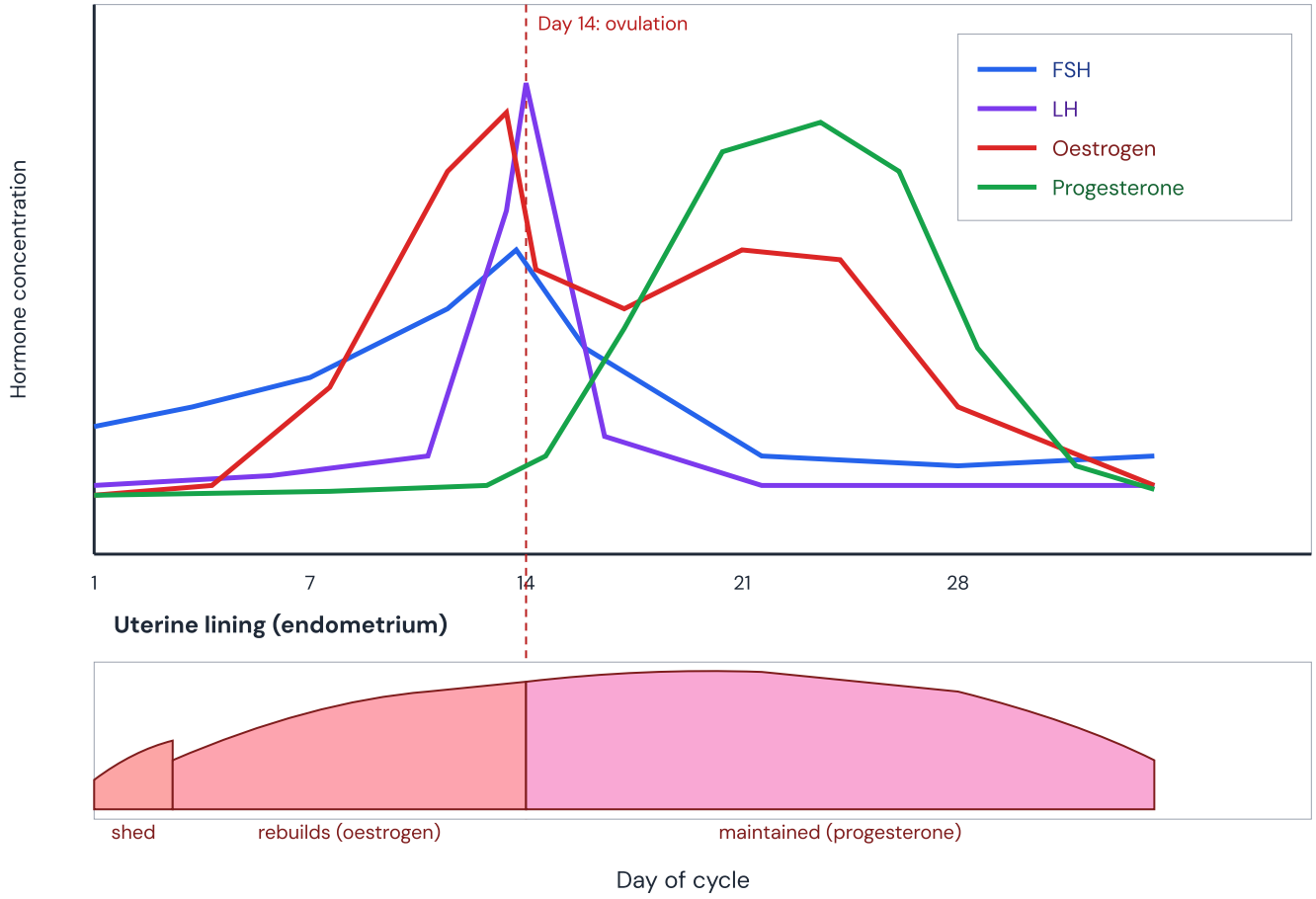
7.6 The Menstrual Cycle

The menstrual cycle is a roughly 28-day cycle controlled by hormones from the pituitary gland and the ovaries. It prepares the body for pregnancy each month. Four main phases:

Days	Phase	What happens
1–5	Menstruation	Uterus lining (endometrium) breaks down and is shed if there is no pregnancy
5–14	Follicular phase	FSH stimulates a follicle in the ovary to develop. The follicle secretes **oestrogen**, which rebuilds the uterus lining

~14	Ovulation	A surge of **LH** (triggered by high oestrogen) causes the follicle to burst and release an egg
14–28	Luteal phase	The empty follicle (now called the **corpus luteum**) secretes **progesterone**, which maintains the uterus lining. If no fertilisation occurs, progesterone falls and menstruation begins again

Hormone Levels and Uterine Lining over the 28-day Menstrual Cycle



7.7 Role of FSH, LH, Oestrogen and Progesterone

Hormone	Origin	Action
FSH (follicle-stimulating hormone)	Pituitary	Stimulates a **follicle** (containing an egg) to develop in the ovary; stimulates the ovary to release oestrogen

Oestrogen	Ovary (follicle)	Rebuilds the uterus lining; inhibits FSH (negative feedback); at very high levels triggers LH surge (positive feedback just before ovulation)
LH (luteinising hormone)	Pituitary	A surge around day 14 triggers **ovulation** and converts the empty follicle into the corpus luteum
Progesterone	Ovary (corpus luteum)	Maintains the uterus lining ready for implantation; inhibits FSH and LH; falling progesterone triggers menstruation

Exam Tip: A common 4-mark question asks you to describe the sequence: "FSH → follicle & oestrogen → high oestrogen triggers LH → LH triggers ovulation → corpus luteum secretes progesterone → progesterone maintains lining." Learning this loop gets you most menstrual cycle marks.

7.8 Hormonal Contraception

Hormonal contraceptives use synthetic oestrogen and/or progesterone to stop the ovary releasing an egg.

Method	Hormone(s)	How it works
Combined pill	Oestrogen + progesterone	Inhibit FSH → no follicle develops; progesterone thickens cervical mucus (blocks sperm)
Progesterone-only pill (mini-pill)	Progesterone	Thickens cervical mucus; thins uterus lining; sometimes stops ovulation
Implant	Progesterone (slow release from rod under skin in arm)	As above; lasts up to 3 years
Injection	Progesterone	As above; lasts ~3 months per injection

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Intra-uterine device (IUD)	Some release progesterone (hormonal IUDs)	Prevent implantation; thicken mucus

Advantages and Disadvantages

- **Advantages:** very effective (>99% when used correctly), reversible, long-lasting options available.
- **Disadvantages:** do not protect against STIs, can have side effects (headaches, weight change, mood changes, small increase in blood clot risk with combined pill), user has to remember to take the pill daily.

7.9 Barrier Methods of Contraception

Barrier methods physically stop sperm reaching the egg.

Method	How it works	STI protection?
Condom (male/female)	Latex sheath physically blocks sperm	Yes — only effective method against most STIs including HIV
Diaphragm / cervical cap	Dome-shaped cap placed over the cervix before intercourse; usually used with spermicide	Partial; not reliable for STIs
Spermicide	Chemical kills sperm; used with another method	No

Surgical methods (**sterilisation** — cutting/tying the oviducts in women or sperm ducts in men) are permanent, highly effective but are not easily reversible. **Natural family planning** (abstaining during fertile days detected by temperature or mucus) has a high failure rate in practice.

7.10 Infertility Treatments

If a couple cannot conceive, hormonal or assisted reproductive treatments can help.

Treatment	How it works	Notes
Clomifene	Blocks oestrogen's negative feedback on the pituitary → pituitary releases more FSH and LH → stimulates ovulation	Used when a woman does not ovulate regularly; can cause multiple ovulations (twins/triplets)
IVF (in-vitro fertilisation)	(1) Woman given FSH & LH to stimulate multiple eggs. (2) Eggs collected from ovaries. (3) Eggs mixed with sperm in a dish — fertilisation occurs outside the body. (4) Embryos grown for a few days. (5) One or two embryos placed into the uterus	Expensive, emotionally demanding; success rate per cycle ~30%; risk of multiple births

Ethical Issues with IVF

- Unused embryos are usually frozen or discarded — some people believe an embryo has the same moral status as a child.
- Multiple births (twins/triplets) carry risks for mother and babies.
- Cost and availability on the NHS are limited — is it fair that some can pay privately?
- Concerns about screening embryos (pre-implantation genetic diagnosis) leading to "designer babies".
- Religious objections to fertilisation occurring outside the body.

7.11 ADH and Water Balance (Osmoregulation)

Osmoregulation is the control of water and ion concentration in the body's cells and blood. Water is gained through drinking and food; it is lost in urine, sweat, breath and faeces. The body must balance these so that the blood plasma has a constant water potential — if it became too dilute or too concentrated, cells could swell and burst or shrink and lose function by osmosis.

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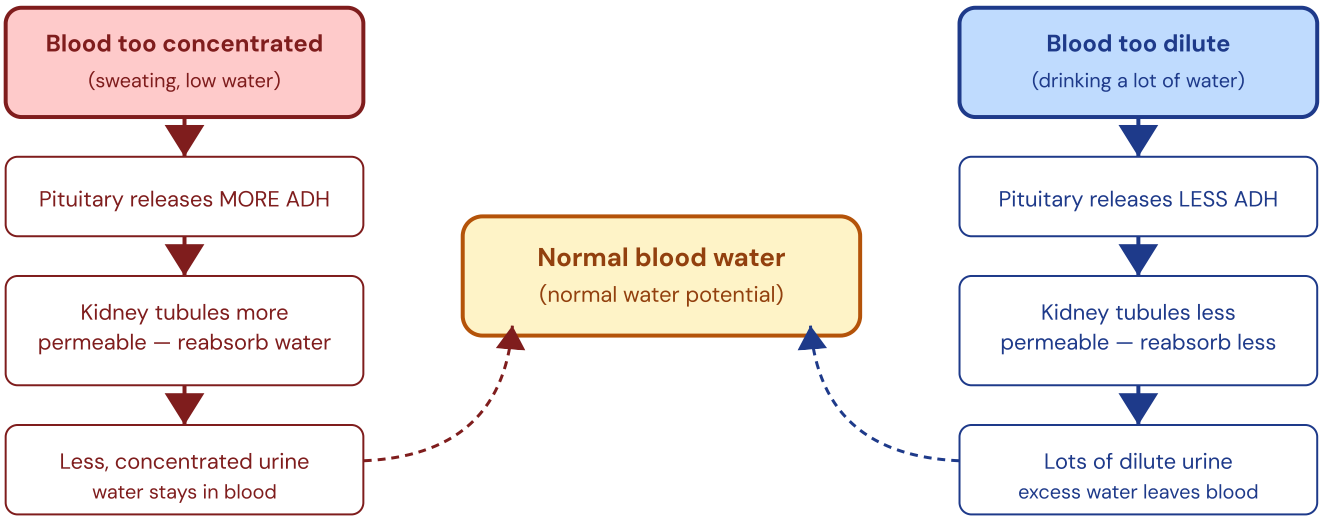
ADH (antidiuretic hormone) is released from the **pituitary gland** when the blood is too concentrated. It acts on the **kidney tubules** (especially the collecting duct and distal tubule), making them **more permeable to water** so more water is reabsorbed into the blood. The result: less, more concentrated urine.

Trigger	ADH level	Kidney response	Urine produced
Blood too concentrated (exercise, sweating, low water intake, salty food)	**High**	Collecting duct very permeable — lots of water reabsorbed	Small volume, concentrated (dark yellow)
Blood too dilute (drinking a lot)	**Low**	Collecting duct less permeable — less water reabsorbed	Large volume, dilute (pale / clear)

ADH — Negative Feedback Water Balance



Each path is negative feedback: the response reverses the initial change and restores the set point.

7.12 The Kidney

The kidneys are bean-shaped organs in the lower back that filter blood to remove waste (urea), regulate water balance and adjust ion concentrations. Each kidney contains about a million filtering units called **nephrons**.

Kidney structure

- **Cortex** — dark outer region where most filtration occurs (contains glomeruli and Bowman's capsules).
- **Medulla** — inner region where loops of Henle and collecting ducts sit (concentrating urine).
- **Pelvis** — central cavity where urine collects before leaving down the ureter.
- **Renal artery** brings blood in; **renal vein** takes cleaned blood out; **ureter** carries urine to the bladder.

The nephron — two key processes

1. **Ultrafiltration** (in Bowman's capsule/glomerulus): high pressure in the glomerular capillaries forces water, glucose, ions, urea and other small molecules out of the blood into the Bowman's capsule. Blood cells and large proteins stay in the blood.
2. **Selective reabsorption** (mainly in the proximal convoluted tubule): all the glucose, most of the water and useful ions are reabsorbed back into the blood. Urea is not reabsorbed.

The **loop of Henle** creates a salt gradient in the medulla so that water can be reabsorbed from the **collecting duct**. ADH controls how permeable the collecting duct is — the final step in adjusting urine concentration.

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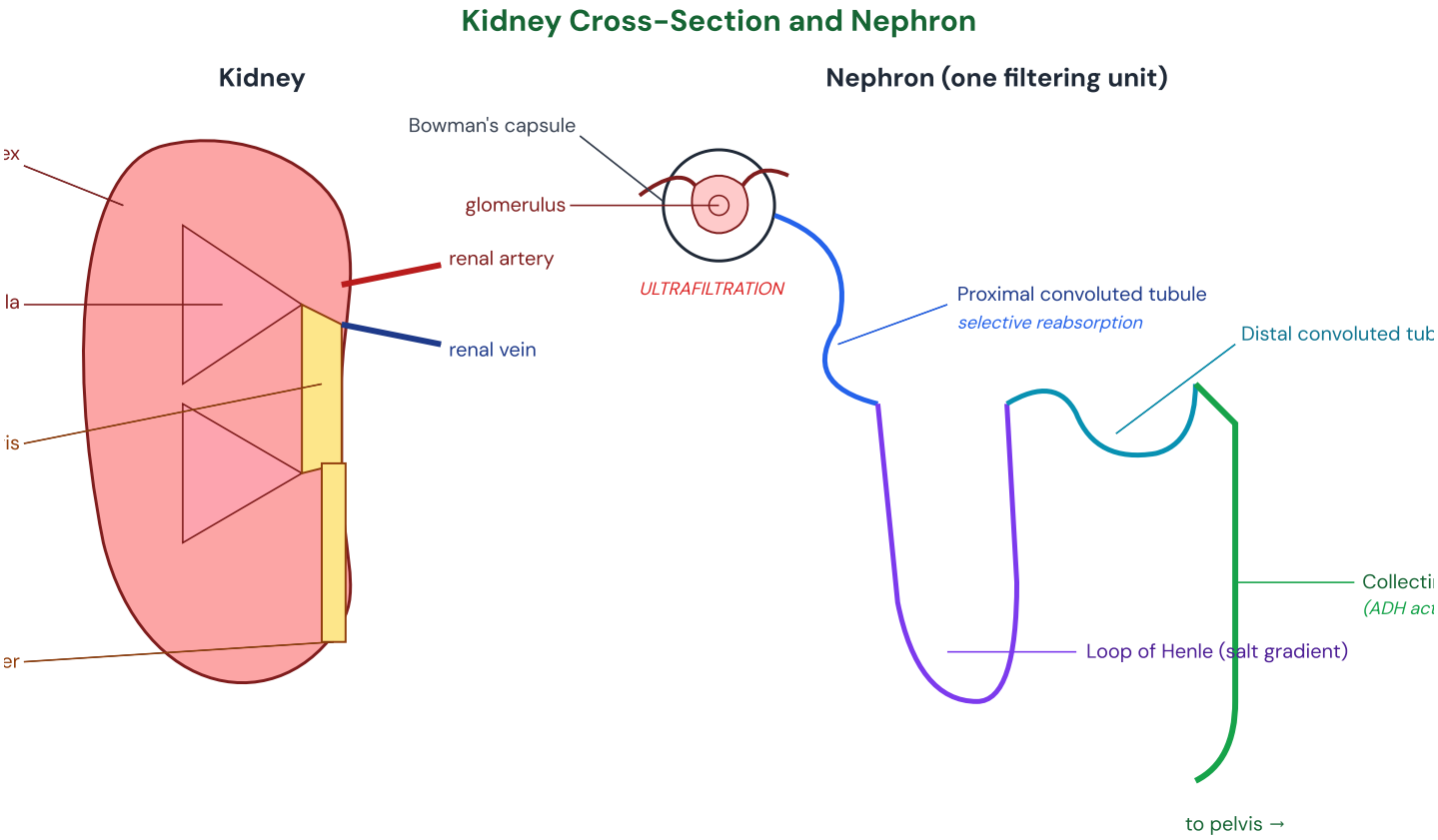

Exam Tip: For a "what is reabsorbed?" question: **all glucose, most water, useful ions** are reabsorbed (selective reabsorption). **Urea is NOT reabsorbed** — that's how we excrete it. Blood cells and big proteins never leave the blood in the first place because they are too large to pass through the filter.

7.13 Urea Formation

Your body cannot store excess amino acids or protein. Instead, the liver breaks down surplus amino acids in a process called **deamination**:

1. The amine group ($-NH_2$) is removed from each surplus amino acid in the liver.
2. The amine group is converted to **ammonia** (NH_3), which is very **toxic**.
3. Ammonia is quickly combined with CO_2 to form **urea**, which is much less toxic but still must be removed.

4. Urea travels in the blood to the kidneys, where it is filtered out in ultrafiltration and excreted in the urine.
5. The remaining carbon-containing part of the amino acid is converted to glucose or fat and used for respiration or stored.

> **Deamination**: the removal of the amine group from an amino acid in the liver, producing ammonia (toxic), which is then converted to urea (less toxic) for excretion by the kidneys.

7.14 Kidney Failure Treatments

If both kidneys fail (because of diabetes, high blood pressure, genetic disease, infection), urea builds up in the blood and ion/water balance is lost. Two treatments are available:

1. Kidney Dialysis

A dialysis machine acts as an artificial kidney. The patient's blood flows along one side of a **partially permeable membrane**; **dialysis fluid** flows on the other side in the opposite direction.

- Dialysis fluid contains **the same concentration of glucose and mineral ions as normal blood**, so these do not diffuse out (no concentration gradient).
- It contains **no urea**, so urea diffuses out of the blood into the dialysis fluid down its concentration gradient.
- Excess ions and water also leave the blood, matching normal blood composition.

Patients usually need dialysis 3 times a week, for around 4 hours each session. It is life-saving but time-consuming and restricts diet.

2. Kidney Transplant

A kidney from a donor (often a close relative or a dead donor) is surgically placed in the patient. The donated kidney takes over full function, ideally for many years.

Treatment	Advantages	Disadvantages

Dialysis	Available quickly; no surgery; no rejection	Time-consuming (~12 hours/week); restricted diet; risk of infection at tube sites; not a cure
Transplant	Long-term cure; free diet; better quality of life	Shortage of donors — long waiting list; surgery risks; must take **immunosuppressants** for life to prevent rejection; immunosuppressants increase infection risk

Exam Tip: Dialysis fluid has the same salt and glucose concentration as blood so these are NOT lost. Only waste (urea) and excess water/salts move out — down concentration gradients. This is a favourite question.

7.15 Thermoregulation (Control of Body Temperature)

Core body temperature must be kept close to 37 °C because enzymes have a narrow optimum. The **hypothalamus** in the brain monitors blood temperature and also receives input from temperature receptors in the skin. It coordinates responses through the nervous system (mostly) and hormones.

Too Hot

- **Sweating** — sweat glands release sweat onto the skin; evaporation of water removes heat (it takes a lot of energy to evaporate water).
- **Vasodilation** — arterioles supplying skin capillaries widen, so more blood flows close to the surface → more heat lost by radiation. Skin appears red.
- **Hairs lie flat** — erector muscles relax; no insulating air layer is trapped.
- Behavioural: removing clothing, drinking cold water, moving to shade.

Too Cold

- **Shivering** — rapid involuntary muscle contractions; respiration in the contracting muscles releases heat.
- **Vasoconstriction** — arterioles supplying skin capillaries narrow; less blood flows near the surface → less heat lost. Skin appears pale.

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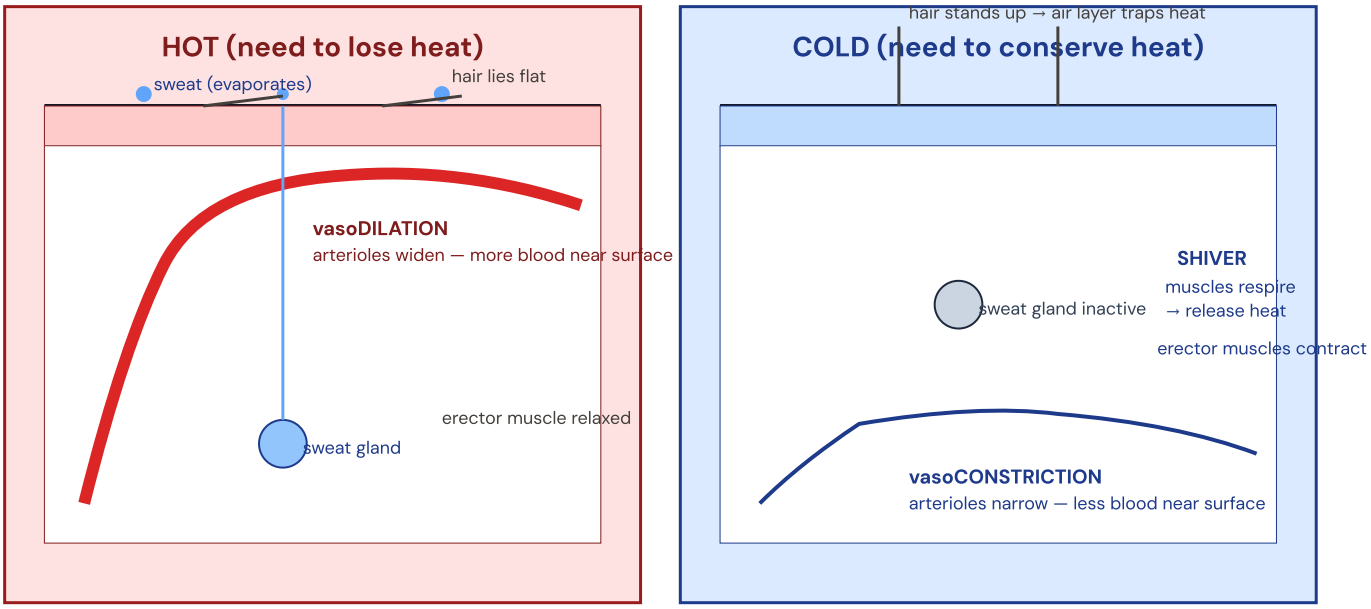
- **Hairs stand up** — erector muscles contract; hairs trap an insulating layer of air. (Not very effective in humans, but visible as "goosebumps".)
- **Reduced sweating**.
- Behavioural: putting on clothes, curling up, drinking hot drinks.

> **Exam Tip:** A common trap: vasodilation does NOT mean blood vessels move closer to the skin. The capillaries are always where they are; the arterioles **widen** so more blood flows through the capillaries near the surface. Write "arterioles dilate so more blood flows through capillaries near the skin surface".

Skin in Hot vs Cold Conditions



Glossary — Topic 7

Term	Definition
Hormone	A chemical messenger secreted by an endocrine gland into the blood; acts on distant target organs.

Endocrine gland	A gland that secretes hormones directly into the bloodstream (no ducts).
Target organ	An organ containing cells with receptors for a specific hormone; only these cells respond.
Pituitary gland	The "master gland" at the base of the brain that releases FSH, LH, ADH and growth hormone, controlling many other glands.
Homeostasis	Maintenance of constant internal conditions in the body.
Negative feedback	A control mechanism in which a change from the set point triggers a response that reverses the change.
Adrenaline	Hormone from the adrenal glands that prepares the body for fight or flight.
Thyroxine	Hormone from the thyroid gland that controls basal metabolic rate.
Insulin	Hormone from pancreatic β-cells that lowers blood glucose by causing liver/muscle uptake and glycogen storage.
Glucagon	Hormone from pancreatic α-cells that raises blood glucose by stimulating glycogen breakdown in the liver.
Glycogen	Storage polymer of glucose found in liver and muscle.
Type 1 diabetes	Condition in which β-cells are destroyed (autoimmune) so little or no insulin is produced. Treated with insulin injections.
Type 2 diabetes	Condition in which body cells become resistant to insulin (linked to obesity). Treated with diet, exercise and drugs.
FSH	Follicle-stimulating hormone from the pituitary; stimulates follicle development and oestrogen release.
LH	Luteinising hormone from the pituitary; surge triggers ovulation.

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Oestrogen	Ovarian hormone that rebuilds the uterine lining and inhibits FSH.
Progesterone	Ovarian hormone from the corpus luteum that maintains the uterine lining.
Ovulation	Release of an egg from the ovary, triggered by an LH surge around day 14.
Clomifene	Drug that blocks oestrogen's negative feedback, raising FSH and LH so the ovary releases an egg.
IVF	In-vitro fertilisation — egg and sperm combined outside the body; embryo placed in uterus.
ADH	Antidiuretic hormone from the pituitary; increases water reabsorption in kidney collecting ducts.
Osmoregulation	Control of the water and solute concentration of body fluids.
Nephron	Functional unit of the kidney; contains Bowman's capsule, proximal tubule, loop of Henle, distal tubule and collecting duct.
Ultrafiltration	High-pressure filtration of blood at the glomerulus/Bowman's capsule; small molecules forced into the nephron.
Selective reabsorption	Reabsorption of useful substances (all glucose, most water, useful ions) from the nephron back into the blood.
Urea	Nitrogenous waste made in the liver from excess amino acids; excreted by the kidney in urine.
Deamination	Removal of the amine group from amino acids in the liver, producing ammonia which is converted to urea.
Dialysis	Treatment for kidney failure in which blood is filtered through a partially permeable membrane against dialysis fluid.

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Vasodilation	Widening of arterioles supplying skin capillaries — more blood flows near the surface, more heat is lost.
Vasoconstriction	Narrowing of arterioles supplying skin capillaries — less blood flows near the surface, less heat is lost.

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